

Bandwidth Swap: A Peer-to-Peer Bandwidth Exchange

Robert Douglas
and Chris Odom

bandwidthswap@pm.me

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admin@bandwidthswap.lol

Abstract. *A marketplace measured in available bandwidth has the possibility of identifying each pipe's market price. When pipe value can be accurately determined, and the marketplace is decentralized with a low barrier to entry, growth and development of infrastructure has the capability to be optimized by the "invisible hand," promoting robustness and consumer needs above all else. We propose improving the growth and development of communication infostructure through peer-to-peer networks enforced by blockchain technology, directly associating incentives for hosts to build and maintain infostructure, with the speed, reliability, and uptime interests of the user.*

Consensus mining has been successfully tested in other markets; our concept, proof-of-bandwidth, can offer utility when the bandwidth consumed to measure a host's proof-of-work is used as a consumer product.

The network distributes rewards to relays in the network based on hop difficulty, when the payload arrives at its intended destination within the allotted timeframe.

This checksum and reward distribution encourages all payloads to be delivered and bad actors to be reported. Each relay's difficulty, determining their transfer fee reward, is calculated by the opportunity cost of an alternative route around that relay. These human operator incentives encourage the construction of redundant and robust networks.

1. Introduction

Communications infostructure growth has been historically retarded by monopolies and exclusion contracts designed to restrict users' optionality such that a user is more likely to be a long-term customer of an ISP service, since ISPs pay a high initial capital expenditure to install in each area. This model protects companies' cash flow, but allows conditions that impede competition and does not promote the lowest price for the user, nor encourage innovation. Existing lines can be used for further redistribution rather than as endpoints,^[i] and many jurisdictions have already provided the legal ease to encourage just this.^[ii]

We propose a fundamental improvement to this supplier-consumer relationship by offering a peer-to-peer network where bandwidth is the product. A blockchain intermediary has the possibility to facilitate rewards and punish poor behavior by making it expensive for a bad actor to lie. When rewards are issued by enabling a customer relay and endpoint service, and there is a bonded barrier of entry to collect those rewards, only decentralized autonomous software is needed to govern the network.

A network capable of rapid growth--one that incentivizes communication innovation--may be the best network for surviving a Carrington event, and offers improved support to industrial, scientific, and medical communications.

2. Blockchain

The \$BANDWIDTH blockchain is designed to offer users autonomy while incentivizing operators at any scale. Built around Ethereum's blockchain and through its ERC-20 token protocol; with Open-Transactions it is designed to offer off-chain settlement for little or no fees, and complex financial instruments^[iii] so

a first of its kind derivative marketplace may be available on launch to promote community scaling.

The \$BANDWIDTH token is the reward issued to Miners by active pool Mints. It is fungible across multiple market makers with different price settlement techniques such as Curve, Balancer, or Uniswap.

Proof-of-bandwidth can be compared to proof-of-work. But rather than having computational machines run arbitrary calculations to bundle a limited number of transactions into a block every 10 minutes, proof-of-bandwidth involves user-supporting nodes known as Miners who announce their status and work every minute to Mints who bundle these announcements into blocks that can be used to measure routes available for users and network health.

3. Minting (Validator)

In Bitcoin, miners solve a cryptographic puzzle, with the first one to do so receiving that block's reward, encouraging people to pool their computing resources, spreading the reward across those who contributed. This promotes a form of centralization^[RD1] in mining pools.

To discourage burdening service operators as well as to encourage integrity and distribution of the data load, we propose two layers of operator infostructure to support the user-operator-blockchain incentivization cycle. Bandwidth Miners, who facilitate user interactions measured in the \$BANDWIDTH token, are given those tokens by \$BANDWIDTH Mints who stake a large number of their own assets and who collect, validate, and publish that interaction data to the (DHT or whatever) \$BANDWIDTH blockchain, receiving a reward for the act.

User software queries for available routes, seeking the information that's aggregated by Mints, significantly reducing the cost of route calculation for users. ...and unburdening them from the ability to discriminate.

- How Miners are compensated for provided transport.
- How Minters are compensated for aggregating route data.

Mints who have staked a large number of assets are compensated by the blockchain through an election process promoting those who have staked more. Every minute, for every block, one Mint is randomly chosen, based on the likelihood of how much they have staked, to validate the next block. This replaces the concept of traditional miners, with validators who, instead of mine a new block, mint new blocks for little computational energy. Should a validator mint a fraudulent block, they will a greater number of assets staked by them than the validation fee they received. Should a validator want their staked assets back, they can after a time greater than is needed to verify they have not previously minted a fraudulent block. The current minimum criteria for being chosen is 1/10,000 of all assets in circulation being staked, and a minimum of 10mb/s of bandwidth.

If the three largest Bitcoin mining pools were to come together they would control more than 51% of the computational power, allowing them certain attacks on future transactions. In comparison, the proof-of-bandwidth protocol promotes a low hardware barrier to entry, encouraging more independent validators, making the network more decentralized and more secure.

To discourage monetary centralization, the age of staked assets is a factor in the selection of validators.

To further reduce the risk of a validator not doing their job, the proof-of-bandwidth algorithm takes five stakers on every block and seeks a minimum of three-out-of-five consensus, reducing the damage caused by a validator not doing their job, and punishing that validator by announcing their address to the blockchain. If their consensus is incomplete, including if the validator is offline, the inconsistent validator gets none of the mined block rewards.

These validators we intend to call Mints, who stake a large number of assets, enabling them to mine new blocks for rewards they are expected to mostly distribute, minting new money for the Miners offering endpoint and relay services.

4. Mining (Hosts)^[RD2]

Miners fulfill the needs of users in the form of Relay or Endpoint services, spending computational energy and bandwidth to facilitate the data requested by users and announcing that fulfillment to Mints for reward.

Endpoint services allow a user to query a known location, and extract information regardless if it is identifiable files by hash or a Web2.0 DataStream. This interaction is similar to a user's relationship with VPNs or UseNet.

Relay services allow users to query an unknown location and have their information securely passed to another relay or endpoint where only the edge devices are known by the user. A relay's minimum price and its minimum transaction fee are determined by the total cost of an approved transaction's routing difficulty, a variable that can be set by a user.

Route subsidy bonuses will be decided by the Network Steward, a DAO designed to receive 10% of every new block reward, decreasing every year by 1% until this project is 9 years old where it'll maintain to help provide the infrastructure for all to become connected. These subsidy bonuses will be offered by the Network Steward defining a clear start point and endpoint, and any nodes that offer an acceptable throughput or lack of delay will be compensated by measuring their opportunity cost against the total path.

A node that has five other acceptable intermediaries in a possible chain will receive $1/5^{\text{th}}$ of that location's reward (example step 1), while a single node connecting a bridge will receive the full reward (example step 4) in maintaining a chain. Route subsidy rewards will be defined by the DAO.

Miners, like Mints, must stake a small portion of assets on the Bandwidth blockchain for a Mint to provision it an Open Transactions NYM (similar to a registration number, but semi-anonymous) that it may broadcast in its header so other devices can know to securely connect to it^[RD3]. Currently, the minimum is computed by taking \$10 USD and measuring against the price of bandwidth on the Stasoshi Dex^[RD4].

An uptime penalty will tarnish a node's ability to earn by deferring that amount to the Network Steward who is responsible for encouraging growth and development of the network. The uptime penalty is measured in seconds of a week with $(\text{number of seconds active}) \div 604,800$ (the total number of seconds in a week) defining the reward that an operator may receive from a user while the difference is sent to the Network Steward. This allows the system to operate normally once a node is turned on, while punishing nodes that turn on just for bonuses and during rate spikes, encouraging a long-life and stable network.

5. Announcements^[RD5]

Miners such as relays and endpoints in a pool of other connected devices announce to a Mint once per minute a 62 byte message containing a string of: (1) their Bandwidth-ID, set by the Open-Transactions NYM (2) their offered difficulty, or the minimum transaction amount they're willing to pass on a message for; (3) their uninterrupted uptime, or how many seconds of all the seconds the device has been active has it been able to talk to a Mint; (4) the nodes maximum available bandwidth pressure whenever it's operator allows a sync-test to run; and (5) the active votes cast by the miner with against the publicly available Network Steward.

[Figure 1]

(00040,000000,1630261675,1630264675,0000000034,0000000037,0000)

1	2	3	3	4	4	5
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This allows the minimum necessary information to be written to the blockchain immutably, while allowing auxiliary services to find and predict the best routes given a user's search criteria, such as uptime or speed, independent reporting the location of the previously registered nodes.

Interestingly Open-Transaction's features of claims and notaries allow a layer two label and verification of a particular node's specific location or trustee should a specific person or group want to announce a claim.

Should an operator see a direct connection to another node greater than can be reached by direct relay, or not currently accepting lookup. Its Open-Transactions NYM lookup code could be offered by the owner over a third-party service such that two connections can be specifically linked on this protocol without outside interference.

Sync-tests are a protocol enforced measurement that is run whenever a Miner establishes, or re-established a connection to a Mint reporting other connected devices broadcasting the \$BANDWIDTH header, and maximum throughput that node is able to offer to the network.

6. Network Steward^[RD6]

The Network Steward, a DAO^[RD7], is purposed is to seed projects without bias and in such a way it promotes healthy competition to grow the \$BANDWIDTH ecosystem. Its goal is to promote competition to keep prices the most competitive for users. The DAO will attempt to encourage a high rate of innovation ^[RD8] across all market segments that use the \$BANDWIDTH protocol.

All proceeds received and spent by the DAO will be reported in the form of annual statements as it will be held to the standard set by Wyoming in the Decentralized Autonomous Organizations Supplement. Tokens unspent after 90 calendar days will be burned and reported to the network. Voting is performed by “marking” a Miner output with specific vote data. Each miner votes though a Minter. These votes are may be proposed by anyone and specifically address the registered the membership, leadership, goals, and spends of the registered Wyoming DAO representing \$BANDWIDTH.

7. Governance

As this system’s deployment takes up physical space unlike software where the space is undefined, it is important that one individual's preferences do not risk injuring or inappropriately change others ability to act in areas they do not affect. Thanks to special tests conducted by RUNE, PBC in 2018, at the time the active developer of Open-Transactions; initial prototypes of location-based notary services that interfaced trustless via the BIP-47 payment privacy protocol, confirm the ability to associate one's vote with a physical location, without double voting. This combination of notary services, BIP47, and a alternative use of Chaumian Cash, can guarantee that in governance voting one’s ballot is both all they receive, blind, and can only be cast once.

This proposed notarizing of ballots available, and specifying of location cast, becomes much more valuable when combined with proof of stake behaviors, that limit whale overruling. We propose protecting the ideas of the community by setting strong limits on the right to vote, such as:

- 1) Minimum use restrictions since the last polling period^[1]; such that one must have spent a minimum amount of \$BANDWIDTH in the ecosystem or operate a minimum number of \$BANDWIDTH producing Nodes.
- 2) Minimum time restrictions since account creation; such that rash, emotional, or undeveloped requests are encouraged to be rejected.
- 3) Anti-private interest restrictions; deployed by logarithmically associating the staking of 1 \$BANDWIDTH as one vote, 10 \$BANDWIDTH as 2 votes, 100 \$BANDWIDTH as 4 votes and 1000 \$BANDWIDTH as 5 votes into continuum.

These restriction have the capacity to protect the ideas of the community while awarding those with more significant capital interest tools for communication and identification without having doctoral control of any particular vote.

These voting capabilities have been previously developed by Voting Matters, a wholly owned subsidiary of MatterFi, the current developers of Open-Transactions. When KYC is capable of restricting whales, consolidated private interests are less likely to win.

^[1] Possible with Pseudonyms through Open Transactions and OBPP-05 (<https://github.com/OpenBitcoinPrivacyProject/rfc/blob/master/obpp-05.mediawiki>)

^[i] https://en.wikipedia.org/wiki/Local-loop_unbundling#United_States

^[ii] <https://web.archive.org/web/20081202153609/http://www.ictregulationtoolkit.org/en/Document.2904.pdf>

[iii] <http://opentransactions.org/wiki/index.php?title=About>

[RD1] those with more powerful equipment, the higher your hash rate, the greater the chance of you'll have to mine the next block and receive the block reward. To increase these chances further miners can come together in pools to combine their computing power and distribute their rewards evenly

[RD2] Unlike the Mints of \$BANDWIDTH, risking their assets in exchange to help be, the Oracle of consensus that rewards operators fulfilling the needs of users, Miners.

[RD3] Perhaps this should be offered by Notary services instead...

[RD4] This is a fictional DEX that will be the first to maintain the exchange of \$BANDWIDTH (to determine a spot price)

[RD5] Should an outside querying the node chose to request **complete status**, the node would report back, the names of other connected nodes, the route bonus currently being collected if a route bonus is available to them

Should an operator allow others, or choose themselves to run a **sync test** (which can be programmatically set at a fixed price for a user to run only if other users are able to see the last time the test was initiated when they query a node) the Relay or Endpoint will scan it's network hardpoints for all other devices that are broadcasting the \$BANDWIDTH prefix and offer to share their distributed hash table of connected devices with the \$BANDWIDTH prefix allowing a device to query up the number of layers to other device it sees fit, currently this variable is three. The most likely use case is when a user establishes a connection it takes a known route, and asks all nodes in the route to preform a sync such that if a less expensive route is available, with a higher or lower mili-second delay.

[RD6] Thanks to Advocate Kathrine Long and Wyoming Governor Mark Godin Wyoming is the first state in America to recognize a DAO LLC and based on the Decentralized Autonomous Organizations Supplement (DAO Supplement), a bill passed by Wyoming's state senate on March 17th of 2021, there is enough clarity of the legal status of the decentralized autonomous organization (DAO) for

operators to construct and rely on them even though the entity that has not yet been contemplated by federal or state legislatures.

Historically one would construct a semi-autonomous entity similar by initiating a PBC (Public Benefit Company) that would allow the entity to specify a public benefit above the fiduciary requirements of a director to its shareholders, followed by BI laws that force human turn over greater than the risk rate of collusion.

Wyoming's new acceptance and guidance protects DAOs from being sued as general partnerships but also solidify the rights of DAOs as legal persons. The registered name will include "DAO," "LAO," or "DAO LLC," and within the Articles of Organization include a "Notice of Restrictions on Duties and Transfers," stating that the DAO Supplement, underlying smart contracts, articles, and operating agreement, if applicable, of a DAO may define, reduce, or eliminate fiduciary duties and may restrict the transfer of ownership interests, withdrawal or resignation from the DAO, return of capital contributions, and dissolution of the DAO. The Articles may define the DAO as either member-managed or algorithmically managed (only legally formable if the underlying smart contracts can be updated, modified, or otherwise upgraded). These Articles must be amended when the DAO's smart contracts have been updated or changed. In Wyoming a DAO LLC does not allow a member to have the organization dissolved for failure to return the member's contribution of capital, and Unless otherwise provided for in the articles, smart contracts, or operating agreement, a withdrawn member forfeits all membership interests in the DAO, including any governance or economic rights.

This new doctrine is likely to yield ruling in the case of contested liability from creditors, challenged through the court system, although case law has yet color in what favor a Judge may lean. Acceptance of the Network Steward as a DAO by the Wyoming courts enables the DAO to use fiat banking for the distribution of rewards and traditional Internal Revenue Service structured 1099 relationships for the distribution of non-token rewards. Given the governance characteristics of this

token, as it relates to pool operators and supply held, non token distributions will allow many freedoms normally prohibited in undefined prosecution of token-reward marketplace as the securities and exchange act of 1933 has not yet been adopted to the cryptocurrency space.

[RD7] Block rewards undergo a recurring decimation, whereby distributions reduce by 10% every 100 days. Each decimation ensures a smooth issuance decay.

[RD8] of each finger of it's determined products by following the same model. 1) Prove prototype and customer acceptance. 2) Grow to critical mass that the customer lifecycle can be defined. 3) Open source all materials and tools that will allow other companies to compete. 4) Transfer assets to SPV known as Company one and offer incentives for liabilities and talent to join company one. 5) Offer seed funding for two other companies to be started to fulfill the product need. This model was well executed prior in the creation of three state owned, yet competing, Chinese airlines whose case studies can be confirmed.